

White paper: notation and terminology across the compendium

2026-07-30 20:10 PDT (second edition; supersedes the 2026-07-29 diagnosis)

A re-audit of mathematical notation and controlled vocabulary in the manuscripts under analysis/report/, after two repair passes. Records what was fixed, what the repairs themselves broke, and what remains.

Method

Sixteen manuscript sources were searched for the symbols carrying the programme's substantive quantities and for the labels naming its designs, architectures, specifications, and studies. The checks are now mechanized as a linter (`notation-lint.pl`) that skips fenced R chunks and `verbatim` blocks, strips `\texttt{}` and backtick spans, and reports violations of the rules in `analysis/report/NOTATION.md`. Every manuscript was re-rendered after each change; counts below are from the sources and from `pdftotext` on the rendered output.

The corpus is smaller than in the first edition. Papers 01 and 02 have since been reorganized: their narrative variants were promoted to `report.Rmd` and the earlier full and slim versions moved to per-paper `archive/` directories. This audit reflects the promoted files.

State after the second pass

The linter reports **clean** for all sixteen manuscripts except one known false positive (a bare `Sx` inside a backtick span that crosses a line break). All sixteen render. The nine inconsistencies of the first edition now stand as follows.

#	Finding (first edition)	State
N1	Drug indicator: D_{it} used for both the binary state and the continuous exposure	Fixed
N2	c_{bm} used for both a correlation and a regression multiplier	Fixed
N3	Half-life quoted in weeks and in days with no conversion	Fixed by units paragraphs
N4	Four symbols for one estimand	Standardized on $\beta_{bm:D}$
N5	Outcome alternates between Y and the code identifier <code>Sx</code>	Fixed in model definitions
N6	Label collisions across architectures, specifications, sweeps, studies	Fixed, after one false start
N7	Two sample-size conventions (per path, total)	Documented, not resolved
N8	Design labels alternate abbreviation and prose	Open (editorial)
N9	Sign convention stated only in one paper	Open (editorial)

What the second pass found that the first pass missed

The specification relabel was incomplete. The first pass renamed the three carryover specifications A1/A2/A3 to M1/M2/M3 in the carryover manuscript and its variants, but four other papers cite those specifications and were not touched. For a day the compendium said M2 in one paper and A2 in four others, which is worse than the original state, because a reader cross-referencing them would conclude they were different objects.

The chosen replacement collided with an existing scheme. The calibration manuscript already uses M0 through M3 for its working-covariance model ladder, and those labels are keys into archived simulation output, not free text. M1/M2/M3 for specifications therefore reintroduced exactly the class of collision the pass existed to remove. The specifications are now **E1/E2/E3**, after E for the exposure regressor that distinguishes them; E is unused anywhere else in the corpus.

The architectures were the deeper problem. A, B, and C are opaque (a reader must memorize which is which) and they are the reason the specification labels collided in the first place. They are now named by the channel that carries the interaction:

Prose	Legends and tables	Data value
the mean-moderation architecture	Mean	<code>mean_moderation</code>
the covariance-moderation architecture	Covariance	<code>mvn</code>
the dual-channel architecture	Dual	<code>combined</code>

This touched 581 label occurrences across 16 files plus the figure drivers. Data values are unchanged, so no simulation was re-run; the figures were regenerated from existing summaries.

A second British-spelling family survived the first pass, because the first pass enumerated word stems rather than patterns: `flavour`, `rigour`, `endeavour`, `labour`, `tumour`. Fixed in four files.

What the repairs themselves broke

Reporting this is the point of a second audit. Three defects were introduced by the first pass and caught here.

Data-side labels were renamed along with display labels. The specification rename reached inside R chunks, changing `method_levels <- c('A1', ...)` and three `filter(spec %in% c('A2', 'A3'))` calls. Factor levels and filters index stored data values; renaming them silently produced empty selections and `NA` rows rather than an error. The first repair was a display-time mapping that kept the stored values at A1/A2/A3, which is where the third pass then found the deeper problem (below).

A display helper returned NA for unmapped inputs. The first version of `spec_display()` indexed a named vector, so the two robust-inference methods (`lme+CR2`, `GEE+MD`) that are not specifications came back `NA` and printed as `NA` in the development-review tables. Now passes unmapped values through.

Three manuscript files were truncated to zero bytes. A shell loop copied each source to a temporary file, transformed it, and moved the result back. When the cloud file provider returned a spurious `ENOENT` for a source that had just been moved by a concurrent reorganization, the `cp` failed but the shell had already created an empty output file, which `mv` then placed over the source. The files were restored from scratchpad backups. Every subsequent script verifies that the source is non-empty before producing output and that the result has not shrunk before writing back. This is a sharper version of the hazard the project guardrails already warn about: the danger is not only in-place editors, but any pipeline whose failure mode is an empty file.

Third pass: retiring the mapping layer

The second pass left the specification labels stored as A1/A2/A3 and displayed as E1/E2/E3, bridged by a `spec_display()` helper applied at each display site. That is precisely the arrangement that had already produced two of the three defects above, and it is fragile for a structural reason: a mapping has to be applied everywhere, it fails silently when a rename reaches one layer and not the other, and it leaves a stored table that cannot be read without the manuscript beside it.

The mapping has therefore been removed rather than documented. The `spec` values were migrated in place: 21 `.rds` files recoded to E1/E2/E3, with the row count and the absence of legacy values verified on read-back for each, and pre-migration copies kept in `analysis/.spec-migration-backup/`. The producing code was updated in the same direction, in both the quoted literals (`'A2'`) and the unquoted named-vector keys (`c(A2 = ...)`), which are easy to miss because they index the same values through different syntax. The display helpers were then deleted from the three manuscripts that carried them.

Correctness was checked by re-running every affected figure driver and manuscript. The development-review document verifies its own headline ranking inline, and reports the same numbers after migration as before (`E2=0.873`, `E1=0.727 -> PASS`), which is the evidence that the recode moved labels and not values.

One mapping is deliberately retained. The `dgp_arch` values (`mvn`, `mean_moderation`, `combined`) are package API arguments, not free labels, so the manuscripts map them to Mean, Covariance and Dual in the ordinary factor-level-to-label way. The distinction worth holding onto is that a label the project owns should be renamed at source, while a label an interface owns should be mapped at the edge.

What remains open

N7, the sample-size convention. One paper reports N per randomization path and others report it in total. The combined-DGP manuscript documents its own instance and defers to a pending boundary re-run. This needs a simulation, not an edit.

N8 and N9 are editorial: design labels still alternate between abbreviation and spelled-out form, and the sign convention is stated in only one paper. Both are best folded into the `rgt` writing pass rather than done mechanically now.

The `rgt` prose remains unwritten in every paper that still carries the three-part scaffold, so a third terminology pass will be needed once the author's text exists. That is the argument for keeping the linter: it converts these two one-off passes into a check that can run on every render.

A caution on scope. This audit verifies internal consistency of symbols and labels. It does not verify that any numerical result is correct, and the label changes were deliberately confined to display so that no stored result was touched. Two papers were promoted to narrative variants during the pass; their archived predecessors were not re-audited and will not match this scheme if they are ever revived.

The reference itself

The complete inventory follows as an appendix: mathematical symbols, the code identifiers and stored data values they correspond to, and the glossary of controlled vocabulary. It is not reproduced here but bound in verbatim from `analysis/report/NOTATION.md`, which is the canonical file the linter checks against. The two therefore cannot drift apart: this PDF is rendered from the audit and that file together.

Canonical notation, identifiers, and glossary

Originally written 2026-07-30 11:57 PDT; expanded to a full reference 2026-07-30 20:20 PDT

This file is the single source of truth for mathematical notation, code identifiers, and controlled vocabulary across the manuscripts under `analysis/report/`. New manuscripts should adopt it rather than re-deriving symbols, and any deviation should be stated explicitly in the deviating paper’s Methods section.

The audit behind it is `analysis/report/whitepaper-notation-audit.md`, whose PDF binds this file in as its appendix so that the two cannot drift apart.

Symbols and identifiers listed here were extracted from the manuscript sources, from R/, from the drivers under `analysis/scripts/`, and from the archived `.rds` summaries.

Part 1. Mathematical symbols

Indices and structure

Symbol	Meaning
i	participant index
t	timepoint index; weekly measurement occasions throughout
j, ij	period or occasion index within participant (papers 04, 06)
N	number of participants. State whether per randomization path or in total ; both conventions are in use
$n_{\text{reps}}, n_{\text{sim}}$	Monte Carlo replicates per cell
k	cycles per participant in design sweeps; also the Weibull shape parameter in the decay family

Outcome and latent components

Symbol	Meaning
Y_{it}	observed symptom score; the modelled outcome (code column <code>Sx</code>)
BL_i	participant-specific baseline level
BR_{it}	biological (pharmacological) response component
PB_{it}	placebo-belief response component
TV_{it}	time-variant natural-history component
ε_{it}	observation-level residual
u_i	participant random intercept
η	expectancy covariate (measured belief), paper 06

Treatment exposure and carryover

Symbol	Meaning
D_{it}	binary drug state: 1 on drug, 0 off drug
$D_{bc,it}$	continuous exposure-decayed drug indicator: 1 on drug, $e^{-\lambda t_{sd}}$ off drug (code column <code>Dbc</code>)
t_{sd}	time since discontinuation; input to the carryover decay (code <code>tsd</code>)
t_{od}	cumulative time on drug; input to the Gompertz response (code <code>tod</code>)
t_{pb}	time-by-placebo-belief accumulation (code <code>tpb</code>)

Symbol	Meaning
$t_{1/2}$	carryover half-life (code <code>carryover_t1half</code>)
λ	carryover decay rate, $\ln 2/t_{1/2}$ (code <code>lambda_cor</code>)
λ_w	Weibull rate, $(\ln 2)^{1/k}/t_{1/2}$

Decay families implemented on the data-generating side: exponential $e^{-\lambda t}$; linear $\max(0, 1 - t/(2t_{1/2}))$; Weibull $e^{-(\lambda_w t)^k}$; power $\max(0, (1 - t/(3t_{1/2}))^p)$.

Biomarker and moderation

Symbol	Meaning
B_i	participant-level pre-treatment biomarker value (code <code>bm</code>)
b_i	standardized biomarker, $(B_i - \bar{B})/s_B$ (code <code>bm_z</code>)
c_{bm}	moderation parameter of the covariance-moderation architecture: on-drug correlation between biomarker and BR . A correlation, bounded in $[-1, 1]$
β_{bm}	moderation parameter of the mean-moderation architecture: dimensionless multiplier scaling the treatment effect by b_i
$c_{bm,A}, c_{bm,B}$	independent mean-channel and covariance-channel weights of the dual-channel architecture
σ_{BR}, σ_{bm}	standard deviations of the response component and of the biomarker

Estimand, inference, and performance

Symbol	Meaning
$\beta_{bm:D}$	the target estimand: the biomarker-by-treatment interaction coefficient (<code>bm:Dbc</code>)
$\beta_{bm}^{BR}, \beta_{bm}^{PB}, \beta_{bm}^{TV}$	component-specific biomarker slopes (paper 06)
$\beta_{bm}^{\text{lumped}}$	the one-component (single-indicator) biomarker slope
β_D	treatment main effect (papers 04, 05)
$\theta, \hat{\theta}$	a generic interaction coefficient; used only where the argument is deliberately not N-of-1 specific (paper 10)
θ_{true}	calibrated true value, $-c_{bm} \sigma_{BR}$
ρ	AR(1) within-factor serial correlation
κ	ratio of mean model-based standard error to empirical standard deviation; $\kappa > 1$ means a conservative test
δ	standardized bias of the test statistic
ν	denominator degrees of freedom
α	nominal significance level; 0.05 throughout
π	power, $\Pr(p < \alpha)$
γ	latent-class gating slope (paper 03); a $BR \times PB$ interaction coefficient (paper 06)
Z_i	unobserved latent class label (paper 03)

The five rules the symbols encode

1. D_{it} is **binary**; $D_{bc,it}$ is **continuous**. These are different objects, and the distinction is the subject of the carryover-specification manuscript. Never use one for the other.

2. c_{bm} **is a correlation**; β_{bm} **is a multiplier**. They are calibrated to a common numeric scale, because the mean-channel shift is applied as $\beta_{bm} b_i \sigma_{BR}$ on the standardized biomarker, so the reference value 0.45 denotes a matched moderation strength under either architecture. The symbols are nonetheless not interchangeable. A paper reporting both architectures at a matched value may use one shared label, but must say so.
3. **Mathematics in model definitions, code identifiers in code contexts.** Write $Y_{ij} = \beta_0 + \beta_D D_{bc,ij} + \dots$ for the model; write `Sx ~ bm + t + Dbc + bm:Dbc` in `\texttt{}` when quoting the R formula. Never mix the two inside one expression.
4. **State the sample-size convention.** N per randomization path and N in total are both in use and are not interchangeable; every Methods section must say which it means.
5. **State the sign convention.** The three components are non-negative *reductions* in symptom severity, so an increase in any component lowers Y . Treatment effects and interaction coefficients are consequently negative; moderation parameters are positive.

Units

Carryover half-life $t_{1/2}$ is quoted in **weeks** in the interaction-focused manuscripts, on the canonical grid $\{0, 0.5, 1.0\}$, and in **days** in the main-effect and test-procedure manuscripts, on a pharmacokinetic scale with a 3-day baseline. Both conventions are retained because each paper's figures are drawn on its own scale. Every paper using days carries an explicit units paragraph giving the conversion (1 week = 7 days). Timepoints t are weekly in both conventions.

Part 2. Code identifiers and stored data values

These are the names as they appear in R/, in the drivers, and in the archived `.rds` summaries.

Where a stored value is also a reporting label, the two are kept **identical**, so that no display-time mapping stands between the data and the page. A mapping layer is a standing hazard: it has to be applied at every display site, it fails silently when a rename reaches one layer and not the other, and it makes a stored table unreadable without the manuscript beside it. Where a stored value is a package API argument (`dgp_architecture = 'mvn'`), it stays as the API defines it and the manuscript maps it to a reporting name, which is the ordinary factor-level-to-label relationship.

Exported package functions

`buildtrialdesign`, `buildSigma`, `generateData`, `censordata`, `lme_analysis`, `generateSimulatedResults`, `validateParameterGrid`, `cumulative`, `modgompertz`, `trajectoryShape`, `shape_logistic`, `shape_hyperbolic_tangent`, `shape_piecewise_linear_breakpoint`, `characterize_carryover`, `analyze_trial_extended`, `print_carryover_summary`, `print_trial_summary`, `plotfactortrajectories`, `PlotModelingResults`, `reknitsimresults`.

Principal arguments

Identifier	Role
<code>modelparam</code>	list: sample size N , moderation c_{bm}
<code>respparam</code>	Gompertz parameters <code>maxr</code> , <code>rate</code> , <code>disp</code> per component
<code>blparam</code>	biomarker and nuisance means and standard deviations
<code>trialdesign</code>	output of <code>buildtrialdesign</code> : <code>timepoints</code> , <code>timeptnames</code> , <code>expectancies</code> , <code>ondrug</code>
<code>dgp_architecture</code>	'mvn', 'mean_moderation', or 'combined'
<code>carryover_t1half</code>	carryover half-life
<code>lambda_cor</code>	decay rate; derived from the half-life when NA
<code>br_family</code>	response-curve family: 'gompertz' and alternatives
<code>br_p2</code> , <code>br_p3</code>	shape parameters of the alternative families
<code>moderation_scaling</code>	'constant' or trajectory-scaled
<code>cached_sigma</code>	pre-built covariance matrix, reused across parallel workers

Identifier	Role
makePositiveDefinite	positive-definiteness repair switch
n_cores	worker count; -1 auto-detects
save_chunks	progressive checkpointing for long runs

Data columns

ptID (participant), Sx (outcome, the Y_{it} above), bm (biomarker), bm_z (standardized biomarker), Dbc (continuous exposure indicator, $D_{bc,it}$), tsd, tod, tpb, t_wk (time in weeks), path (randomization path).

Stored factor levels and their reporting labels

Column	Stored values	Reported as	Mapped?
spec	E1, E2, E3	E1, E2, E3	no, identical
carryover_form	exponential, linear, weibull	as stored	no, identical
design	CO, Hybrid, 0LBDC	CO, Hybrid, OL+BDC	only 0LBDC to OL+BDC
dgp_arch	mean_moderation, mvn, combined	Mean, Covariance, Dual	yes; these are package API values

The spec column stored A1, A2, A3 until 2026-07-31. Twenty-one .rds files were migrated to E1/E2/E3 and the producing drivers updated to emit them, so the display-time mapping that previously bridged the two has been retired. Pre-migration copies are in analysis/.spec-migration-backup/. Output archived before that date, including anything under a paper’s own archive/, still uses the old values.

Summary-table columns

n_reps, n_converged, converged_frac, non_convergence_rate, power, bias, mean_estimate, true_value, empirical_se, mse, coverage, and an mc_se_-prefixed Monte Carlo standard error for each (mc_se_power, mc_se_bias, mc_se_mse, mc_se_coverage, mc_se_non_convergence).

Part 3. Controlled vocabulary and glossary

The project lexicon at docs/29-nof1-precision-medicine-lexicon.md holds the full terminology set (241 terms across 13 thematic sections, with cross-field synonyms). The entries below are the subset carrying a fixed, enforced meaning in this compendium.

Trial designs

OL (open-label). Unblinded titration; no off-drug contrast, so it supplies no within-subject interaction information. **BDC** (blinded discontinuation). A blinded switch to placebo after open-label response. **OL+BDC**. Open-label titration followed by blinded discontinuation. **CO** (crossover). Traditional two-period within-subject crossover. **Hybrid**. The Hendrickson (2020) design: open-label titration, blinded discontinuation, then a brief crossover. The programme’s reference design. Capitalize it. **Aggregated N-of-1 trial**. Many single-patient on/off series pooled into one mixed model.

Define each abbreviation at first use in every paper.

Data-generating architectures

Named by the channel that carries the interaction. Single letters are no longer used.

Name in prose	Legends and tables	Data value	Former label
the mean-moderation architecture	Mean	<code>mean_moderation</code>	Architecture A
the covariance-moderation architecture	Covariance	<code>mvn</code>	Architecture B
the dual-channel architecture	Dual	<code>combined</code>	Architecture C

Mean-moderation. The biomarker scales the treatment effect in the conditional mean; the interaction lives in the first moment. The near-universal convention in the trial-simulation literature. **Covariance-moderation.** The biomarker-response correlation is treatment-state dependent; the interaction lives in the second moment. The Hendrickson formulation, and the one whose power is most eroded by carryover. **Dual-channel.** Both channels active with independent weights $c_{bm,A}$ and $c_{bm,B}$; the other two are its single-channel limits. A data-generating construct only: the two weights are not separately identifiable from a fitted `bm:Dbc` coefficient.

Analysis specifications for carryover

E1. Binary on-drug indicator; carryover ignored. **E2.** Exposure-weighted continuous predictor D_{bc} , committing to an assumed decay form and half-life. **E3.** Binary indicator plus a lagged just-off-drug nuisance term; the classical Jones-Kenward crossover device, assuming no decay form.

The letter E stands for the exposure regressor that distinguishes them. These were labeled A1/A2/A3 until 2026-07-31; the stored data and the drivers were migrated to E1/E2/E3, so the labels now agree everywhere and nothing is mapped at display time.

Why not A/B/C and M1/M2/M3. The original scheme used A/B/C for architectures and A1/A2/A3 for specifications, colliding on the letter A within a single manuscript. Relabeling the specifications M1/M2/M3 removed that collision but created a second one: the calibration manuscript already uses M0-M3 for its working-covariance model ladder, and those labels are keys into archived data. E1/E2/E3 is free across the whole corpus, and naming the architectures by mechanism removes the class of collision entirely.

Response-curve families

Modified Gompertz (default), $y = \text{maxr} \cdot \exp(-\text{disp} \cdot \exp(-\text{rate} \cdot t))$ with a vertical offset so the curve passes through the origin. Alternatives: **logistic** (symmetric sigmoidal), **hyperbolic tangent** (fast-saturating), **piecewise-linear breakpoint** (no smoothness assumption).

Missing data and dropout

MCAR, MAR, MNAR in the standard Rubin sense. **Informative dropout.** Hazard depending on cumulative symptom worsening since baseline. **Happy accident.** The randomization-path selection effect by which informative dropout preferentially preserves the most informative crossover blocks.

Inference

LME. Linear mixed-effects model; the programme’s default analysis. **corCAR1.** Continuous-time AR(1) residual correlation structure. **GEE.** Generalized estimating equations; the marginal comparator. **Mancl-DeRouen, CR2.** Small-sample bias-corrected sandwich variance estimators. **RM-ANOVA.** Repeated-measures analysis of variance; the strict form requires a dichotomized biomarker. **Anti-conservative.** Realized Type I error above nominal. **Working covariance.** The assumed residual covariance of the LME; reserve **working correlation structure** for the GEE object.

Estimands and reporting

Biomarker-treatment interaction. The programme’s core estimand; equivalently the patient-by-treatment interaction, or heterogeneity of treatment effects (HTE) in the precision-medicine literature. **Predictive biomarker.** Predicts differential benefit, as opposed to a **prognostic** biomarker, which predicts outcome

regardless of treatment. **ADEMP.** The Morris, White and Crowther (2019) reporting framework: aims, data-generating mechanisms, estimands, methods, performance measures. **MCSE.** Monte Carlo standard error; reported with every performance measure.

Enumeration conventions

Simulation studies. Number them (Study 1, Study 2, ...) rather than lettering them. **Sensitivity sweeps.** Labels S1, S2, ... are paper-local and are not comparable across papers. Cite them with the paper prefix when crossing manuscripts (05-S2, not S2).

Prose terminology

Symbol conventions are settled here; prose terminology is settled by `docs/30-terminology-consistency-plan.md`, executed 2026-07-30 with its D1 spelling decision locked to **US English**, overriding that document's original frequency-based recommendation of British forms. The two are complementary: this file governs mathematics and identifiers, that one governs surface forms and acronym expansion.